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ORIGINAL RESEARCH

Individual Variability Is More Important Than Analytical Methods When Calculating Relative Speed of Beverage Bioavailability

Edward M. Balog, Mateo Golloshi, HyunGyu Suh, and Melinda Millard-Stafford
School of Biological Sciences, Georgia Institute of Technology, Atlanta, GA, USA

Deuterium oxide (D_2O) appearance in blood is a marker of fluid bioavailability. However, whether biomarker robustness (e.g., relative fluid delivery speed) is consistent across analytical methods (e.g., cavity ring-down spectroscopy) remains unclear. Fourteen men ingested fluid (6 ml/kg body mass) containing 0.15 g/kg D_2O followed by 45 min blood sampling. Plasma (D_2O) was detected ($n = 8$) by the following: isotope-ratio mass spectrometry after vapor equilibration (IRMS-equilibrated water) or distillation (IRMS-plasma) and cavity ring-down spectroscopy. Two models calculated D_2O halftime to peak ($t_{1/2max}$): sigmoid curve fit versus asymmetric triangle (TRI). Background (D_2O) differed ($p < .001$, $\eta^2 = .98$) among IRMS-equilibrated water, IRMS-plasma, and cavity ring-down spectroscopy (152.2 ± 0.8, 147.2 ± 1.5, and 137.7 ± 2.2 ppm), but did not influence ($p > .05$) D_2O appearance (Δppm), time to peak, or $t_{1/2max}$. Stratifying participants based on mean $t_{1/2max}$ (12 min) into "slow" versus "fast" subgroups resulted in a 5.8 min difference ($p < .001$, $\eta^2 = .73$). Significant $t_{1/2max}$ model ($p = .01$, $\eta^2 = .44$) and Model × Speed Subgroup interaction ($p = .005$, $\eta^2 = .50$) effects were observed. Bias between TRI and sigmoid curve fit increased with $t_{1/2max}$ speed: no difference ($p = .75$) for fast (9.0 min vs. 9.2 min, respectively) but greater $t_{1/2max}$ ($p = .001$) with TRI for the slow subgroup (16.1 min vs. 13.7 min). Fluid bioavailability markers are less influenced by which laboratory method is used to measure D_2O as compared with the individual variability effects that influence models for calculating $t_{1/2max}$. Thus, TRI model may not be appropriate for individuals with slow fluid delivery speeds.

***Keywords**:* stable isotope, cavity ring down spectroscopy, isotope ratio mass spectrometry, fluid delivery

Deuterium oxide (D_2O) is widely used for metabolic applications in nutritional science. One application described decades ago (Davis et al., 1987) used D_2O to compare effects of beverage formulae on relative fluid delivery rates into the blood. Since then, much effort has focused on optimizing oral solutions (Davis et al., 1988, 1990; Jeukendrup & Moseley, 2010; Jeukendrup et al., 2009; Koulmann et al., 1997) to speed the appearance of D_2O into body fluids (Armstrong, Klau et al., 2010; Church et al., 2017).

Early analytical techniques required methods (Davis et al., 1987; Lukaski & Johnson, 1985) to isolate D_2O from interfering components (e.g., proteins and metabolites) in plasma before subsequently measuring an equilibrated water sample with infrared spectroscopy. However, because infrared spectroscopy is less reliable compared with isotope-ratio mass spectrometry (IRMS) with coefficient of variations (CVs) of 4.8% and <2%, respectively (Leiper, 2015), IRMS became the criterion. Recent availability of laser absorption spectroscopy such as cavity ring-down spectroscopy (CRDS) provides another option (dos Santos et al., 2019). CRDS has the practical advantages of requiring less complex sample handling and is not analyst dependent. However, CRDS runs the risk (when measuring plasma or urine) of residual memory errors between successive injections of specimens (Sagayama et al., 2021; Thorsen et al., 2011). When comparing isotope abundance in urine by CRDS and IRMS for total body water analyses (Thorsen et al., 2011), accuracy and precision were comparable ($0.5 \pm 1\%$). To our knowledge, the use of ultrasensitive laser spectroscopy for plasma-based applications is not documented in the nutrition literature. Given the additional requirements to process plasma for IRMS, it is unclear whether utilizing CRDS produces similar results for key blood biomarkers of fluid delivery.

The appearance of D_2O in blood is a balance between rates of entry from the gut and exit into interstitial/intracellular compartments. After ingesting D_2O, there is a rapid rise followed by a more prolonged plateau and/or decline. Typically, the rising phase can be fit by sigmoid curve and time to reach half-maximal [D_2O] ($t_{1/2max}$) calculated. Reports use $t_{1/2max}$ as a marker for relative speed of fluid delivery (Currell et al., 2008; Jeukendrup et al., 2009; Maughan et al., 1990, 2004), but calculation methods are seldom detailed. Sigmoid curve fitting (SIG) allows derivation of $t_{1/2max}$ but lacks a physiological basis. Asymmetric triangle (TRI) is a more complex mathematical model with physiological underpinnings initially derived from gastric emptying studies (Bluck et al., 2002) and later applied to plasma D_2O appearance (Hill et al., 2004).

Therefore, one objective was to compare CRDS with criterion IRMS methods for oral hydration parameters. Despite plasma preparation differences for IRMS, it was hypothesized CRDS and IRMS would have good agreement (minimal bias) for biomarkers of fluid bioavailability. Since Leiper (2015) suggested some measures (e.g., area under the curve) do not provide a measure of speed, our second purpose was to compare two analytical models (SIG vs. TRI) to calculate speed of fluid delivery to determine whether the two approaches yield similar results.

Balog https://orcid.org/0000-0002-2127-9659
Golloshi https://orcid.org/0000-0002-5192-8402
Suh https://orcid.org/0000-0002-4549-5196
Millard-Stafford (mindy.millardstafford@ap.gatech.edu) is corresponding author, https://orcid.org/0000-0003-2944-1494

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Methods

Participants

Fourteen healthy adult men participated in this study, recruited from the local university population. Height, weight, and body composition using seven skinfolds (Jackson & Pollock, 1978) were assessed. Mean ± *SD* physical characteristics were as follows: age 24.1 ± 4.3 years, height 1.82 ± 0.05 m, body mass 74.9 ± 7.3 kg, and body fat 10.8 ± 5.1%. Participants with history of smoking, use of nicotine-containing products, illegal/recreational drugs, medications altering fluid balance (e.g., diuretics), over-the-counter drugs (e.g., antihistamines and stimulants), or chronic digestive disorders/acute digestive illness were excluded. All participants gave written informed consent as approved by the Institutional Review Board.

Test Protocol

Participants reported following overnight fast after hydrating well the evening prior (by consuming an additional 1 L of water above their typical fluid intake and refraining from strenuous exercise). Urine specific gravity was measured by refractometer (URC-Ne Refractometer, Atago) on a first morning catch specimen obtained prior to arrival. Mean ± *SD* urine specific gravity (1.021 ± 0.007) was consistent with normal hydration status (Armstrong, Pumerantz et al., 2010).

After baseline blood sampling, 0.15 g D_2O/kg body mass was ingested by adding to 6 ml/kg body mass of hypotonic electrolyte-only beverage (Powerade Zero, Coca-Cola Company). Mean D_2O dose was 11.4 ± 0.8 g, and fluid volume was 453 ± 28 ml. D_2O was 99.9% enriched (Sigma-Aldrich). The last 50 ml of beverage was used to rinse the cup and ensure complete ingestion of D_2O. D_2O measures were computed as Δppm (subtracting the background baseline [D_2O] prior to drink administration). Blood (6 ml) was obtained from a venous catheter kept patent with isotonic saline and sampled at 2.5, 5, 10, 15, 20, 30, and 45 min into K_3EDTA vacutainers (Becton-Dickinson) and centrifuged at 3,000 rpm. Plasma was stored at −70 °C until samples were either prepared for analyses or shipped in dry ice.

Isotope-Ratio Mass Spectrometry

Samples were shipped as water (IRMS-WT) or plasma (IRMS-PLA) for D_2O analysis. For IRMS-WT, our laboratory performed D_2O separation from plasma by vapor equilibration as initially described (Davis et al., 1987). Plasma (1.0 ml) was added to the inner chamber of diffusion dishes (Bel-Air Products), and an equal volume of deionized tap water was added to the outer chamber. After 48 hr incubation at 37 °C, the outer chamber water was pipetted into crimp-sealed 1.5 ml glass vials (Thermo Fisher Scientific), kept chilled (not frozen), and shipped to Center for Applied Isotope Studies, Athens, GA.

This laboratory was experienced using IRMS to analyze D_2O previously (Armstrong, Klau et al., 2010; Millard-Stafford et al., 2007). The isotopic composition of water (standard or sample) was determined in duplicate using a Thermo Finnigan Delta V Plus IRMS coupled to Thermo Finnigan High-Temperature Conversion Elemental Analyzer via a ConFlo III Universal Interface. A CTC A200S Autosampler was used to inject samples into the Thermo Finnigan High-Temperature Conversion Elemental Analyzer for analysis. All measurements were performed by the same experienced investigator.

Frozen plasma for IRMS-PLA was shipped to Pennington Biomedical Research Center (Clinical Chemistry Core Laboratory) for D_2O determination utilizing a different procedure to distill D_2O from interfering plasma components. Plasma (100 µl) was placed in the inner well of inverted cryovial caps and vials screwed into place. The inverted vial was placed, lid down, into a bed of glass beads heated to 70 °C on a hot plate. Condensate on the cool vial wall was collected, and samples were diluted in 1:3 (distillate: 40 µl + tap water: 120 µl). Samples were analyzed using a ThermoFisher Delta XP Isotope-Ratio Mass Spectrometer. This laboratory previously measured deuterium in doubly labeled water studies (Johannsen et al., 2012).

Cavity Ring-Down Spectroscopy

Plasma samples were analyzed for D_2O by CRDS (Berman et al., 2012) using a Liquid Water Isotope Analyzer with automated injection system, version 2 upgrade (Los Gatos Research) at Metabolic Solutions Inc., Nashua, NH. This laboratory has performed CRDS in other nutrition studies (Damas et al., 2016). Plasma proteins were removed by adding 5 mg zinc sulfate monohydrate to 25–50 µl plasma. Samples were vortexed and spun at 8,000 rpm to precipitate proteins. The plasma protein-free supernatant was injected eight times, averaging the last three measurements. This approach reduces inherent "memory" effects of prior sample injections (Penna et al., 2012). Standard curves were run before and after samples for calculation of D_2O enrichment as Δppm relative to Vienna Standard Mean Ocean Water. Within laboratory precision for low to high [D_2O] standards was 0.9%–1.5%.

Data were measured relative to Vienna Standard Mean Ocean Water and converted to atom% D_2O and ppm (McKinney et al., 1950; Péronnet et al., 2012). Although equations differ slightly among laboratories, in practice, the magnitude of such an effect is minimal (between ~0.7 and 1.3 ppm).

Derivations of Halftime to Maximum

The time course of D_2O appearance in blood was fit by a SIG model with a modified version of an exponential rise to maximum commonly used (Currell et al., 2008; Jeukendrup et al., 2009),

$$\Delta\text{ppm} = \Delta\text{ppm}_{\max}\left(1 - e^{-kt}\right)^c, \tag{1}$$

where t is time (in minutes), $\Delta\text{ppm}_{\max}$ is maximal Δppm, k is appearance rate constant, and exponent c allows for delay in D_2O appearance. The $t_{1/2\max}$ for D_2O appearance, derived from Equation (1) is:

$$t_{1/2\ \max} = \frac{\ln\left(1 - \sqrt[c]{0.5}\right)}{-k}. \tag{2}$$

Nonlinear regression was performed using SigmaPlot 13.0 (Systat Software Inc.).

The TRI model was derived to study gastric emptying (Bluck et al., 2002) and applied to water absorption (Hill et al., 2004). The amount of beverage absorbed as a function of time is best described by a sigmoid curve (Hill et al., 2004); thus, the derivative with respect to time is the rate of absorption. The derivative of Δppm with respect to time was computed, and a plot of time versus derivative was generated using OriginPro Graphing software (version 2021b). For the first and last points (time $t = 0$ and 45), the equation used to calculate the derivative was

$$f(x_2) = \frac{y_2 - y_1}{x_2 - x_1}.$$

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For the time points between 0 and 45, the central difference was calculated by Equation (3) as time points were unevenly spaced.

$$f'(x_i) = \frac{1}{2}\left(\frac{y_{i+1} - y_i}{x_{i+1} - x_i} + \frac{y_i - y_{i-1}}{x_i - x_{i-1}}\right). \quad (3)$$

In practice, two derivatives (T_1 and T_2) are calculated from the time course of blood D_2O accumulation: (T_1) time at the peak derivative (i.e., when the maximal rate of D_2O appearance occurs), and (T_2) time when no further change is occurring; thus, the derivative equals zero. If T_2 did not reach zero at any point, the lowest derivative closer to zero was used. These two parameters (T_1 and T_2) were identified and used to calculate $t_{1/2\max}$ according to:

$$t_{1/2\max} = \sqrt{T_1\left(\frac{T_2}{2}\right)}. \quad (4)$$

Statistical Analysis

Biomarker variables across three laboratory methods were compared using analysis of variance with repeated measures in eight participants. Time course of plasma D_2O and $t_{1/2\max}$ models among laboratory methods were analyzed using two-factor analysis of variance with repeated measures and Holm–Sidak post hoc analysis (SPSS, version 26). Bland–Altman plots (Bland & Altman, 1986) were used to illustrate limits of agreement and mean bias among laboratory methods ($n = 8$) and $t_{1/2\max}$ models ($n = 14$). Linear regression across methods and models was performed using SigmaPlot 13.0 (Systat). The level of significance was set at $p < .05$. Eta^2 (η^2) indicated the level of variance attributed to a factor, with values >.2 typically considered large (Lenhard & Lenhard, 2016). Data are expressed as mean ± *SD*.

An a priori power analysis based on method comparisons was not possible due to the lack of relevant data in the literature. Previous nutritional studies (Jeukendrup et al., 2009), using similarly sized samples ($n = 10$), were sufficiently powered to detect differences in $t_{1/2\max}$ between beverages containing 3% and 6% glucose (6 min mean difference).

Results

Background Plasma D_2O

Prior to D_2O ingestion, background plasma [D_2O] differed ($F = 321$, $df = 2$, $p < .001$, $\eta^2 = .98$) across methods (Table 1). CRDS was lower ($p < .001$) versus IRMS by 9.5–14.5 ppm, but the two IRMS also differed ($p < .001$) by 5 ppm. CVs obtained from each independent laboratory on baseline [D_2O] were 1.6%, 1.0%, and 0.5% for CRDS, IRMS-PLA, and IRMS-WT, respectively.

Peak Plasma D_2O, Time to Peak, and Rate of Accumulation

Peak D_2O was not different ($F = 1.7$, $df = 2$, $p = .23$, $\eta^2 = .19$) among methods (~200 Δppm, Table 1). Time to reach peak D_2O was not different ($F = 1.0$, $df = 2$, $p = .35$, $\eta^2 = .12$) across methods. Across all methods, two "fast" individuals reached absolute peak values at 20 and 30 min but others reached maximum at 45 min. There was no statistical difference among laboratory methods in area under the curve, average, or maximal accumulation rates ($p = .27$, $\eta^2 = .19$; $p = .18$, $\eta^2 = .23$; $p = .63$, $\eta^2 = .06$, respectively).

Agreement Among D_2O Methods

Plasma D_2O Δppm across methods was highly correlated (Figure 1) and met acceptable limits of agreement (Giavarina, 2015). Linear regression of IRMS-WT against CRDS had R^2 of .93 (slope = 0.96) and mean bias of −8 Δppm with 93% of the error falling within 1.96 *SD* of the mean (Figure 1a). Linear regression of IRMS-PLA against CRDS had R^2 of .93 (slope = 0.99) and mean bias of −12 Δppm with 95% of the error falling within 1.96 *SD* of the mean (Figure 1b). Linear regression of IRMS-WT against IRMS-PLA had R^2 of .99 (slope = 1.03) and mean bias of −4 Δppm with 98% of the error falling within 1.96 *SD* of the mean (Figure 1c).

Plasma D_2O Appearance

Interindividual variation in D_2O appearance (Figure 2a–2c) contributed to a relatively large *SD* despite similar group means over time (Figure 2d). Mean plasma D_2O appearance followed a sigmoid time course (Figure 2d) with time effects ($p < .001$) but

Table 1 Mean ± *SD* of Fluid Bioavailability Measures Based Upon D_2O Enrichment ($n = 8$ Participants) Across Laboratory Methods

	CRDS-PLA	IRMS-PLA	IRMS-WT
Background D_2O (ppm)	137.7 ± 2.2	147.2 ± 1.5*	152.2 ± 0.8*,†
δ% vs. VSMOW	−116 ± 14	−55 ± 10*	−18 ± 5*,†
Peak D_2O (Δppm)	193.2 ± 83.3	211.4 ± 75.1	209.3 ± 93.3
Time to peak (min)	39.4 ± 11.2	40.0 ± 9.6	40.0 ± 9.4
Average rate D_2O accumulation (Δppm/min)	5.6 ± 4.6	6.2 ± 4.6	6.2 ± 5.0
Maximum rate D_2O accumulation (Δppm/min)	14.3 ± 10.0	13.8 ± 10.1	13.6 ± 9.4
AUC (Δppm/min)	5,633 ± 3,261	6,223 ± 3,069	6,142 ± 2,598
$t_{1/2\max}$—Asymmetric triangle (min)	12.2 ± 5.0	11.6 ± 4.9	11.4 ± 4.8
$t_{1/2\max}$—Sigmoid curve fit (min)	11.6 ± 3.7	11.8 ± 3.5	11.8 ± 3.3

Note. AUC = area under the curve; CRDS = cavity ring-down spectroscopy; D_2O = deuterium oxide; IRMS = isotope-ratio mass spectrometry; $t_{1/2\max}$ = time to half-maximal; VSMOW = Vienna Standard Mean Ocean Water; PLA = plasma; WT = equilibrated water.
*Difference ($p < .05$) from CRDS. †Difference ($p < .05$) from IRMS-PLA.

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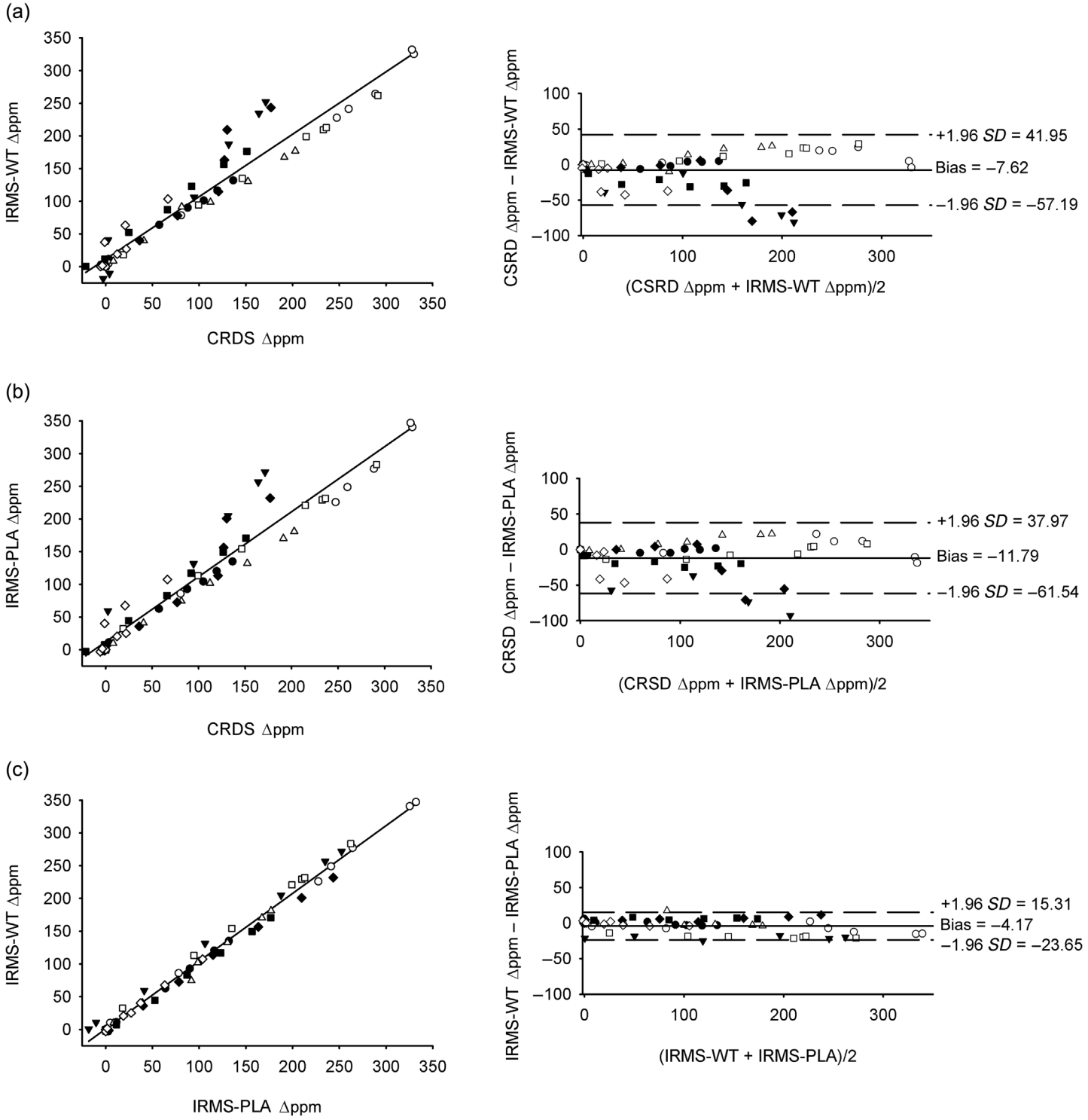

Figure 1 — Regression and Bland–Altman analysis of delta plasma D_2O appearance (Δppm) obtained by CRDS and two IRMS methods. Symbols represent data from individual participants. (a–c) Left panels are the data from one D_2O detection method regressed against a second. The line is the best-fit linear regression (CRDS vs. IRMS-WT: intercept = 11.69, slope = 0.96, r^2 = .93; CRDS vs. IRMS-PLA: intercept = 12.26, slope = 0.99, r^2 = .93; IRMS-PLA vs. IRMS-WT: intercept = 0.71, slope = 1.03, r^2 = .99). The right panels are corresponding Bland–Altman plots illustrating the agreement between the two methods where the solid line represents the mean bias and the dashed lines represent mean error ± 1.96 *SD*. D_2O = deuterium oxide; CRDS = cavity ring-down spectroscopy; IRMS = isotope-ratio mass spectrometry; WT = equilibrated water; PLA = plasma.

no D_2O method ($F = 2.01$, $df = 2$, $p = .18$, $\eta^2 = .22$) or Method × Time interaction ($F = 0.97$, $df = 14$, $p = .48$, $\eta^2 = .12$) effects. Plasma D_2O was elevated above baseline ($p < .05$) by 5 min but values plateaued after 15 min ($p > .05$).

Halftime to Peak ($t_{1/2max}$)

The analytical model used for $t_{1/2max}$ (Table 1) resulted in no difference ($F = 0.146$, $df = 2$, $p = .86$, $\eta^2 = .024$) across laboratory

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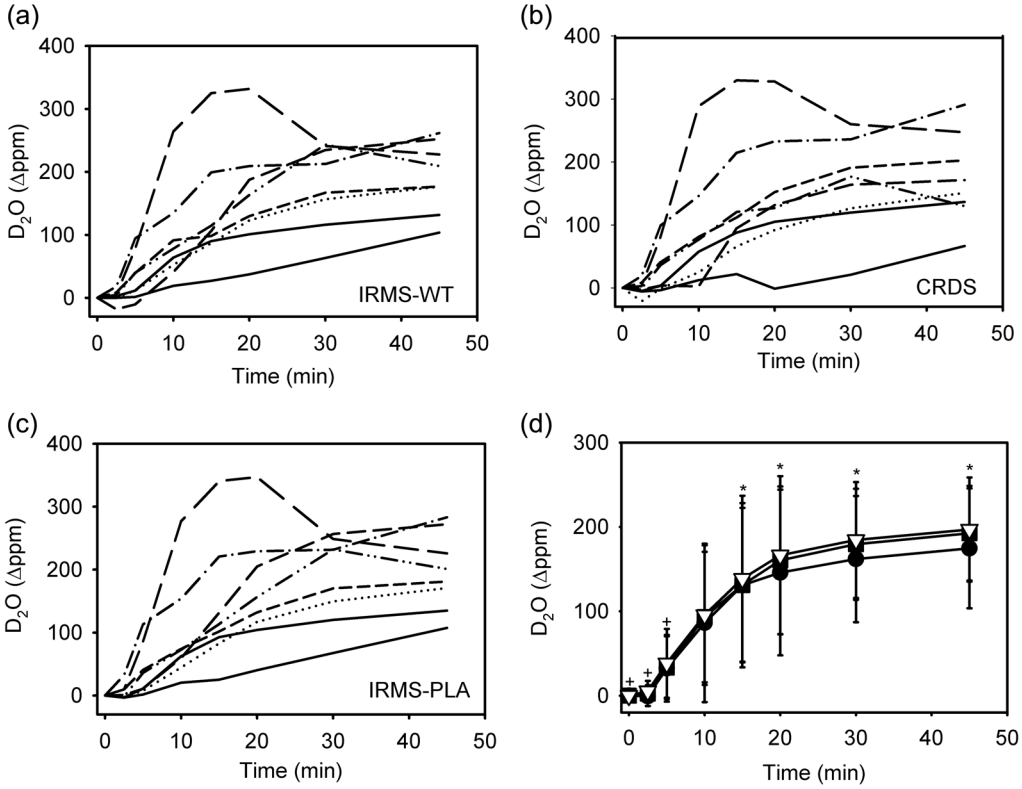

Figure 2 — Time course of D_2O appearance. Plasma D_2O appearance (Δppm) for individual participants as determined by each of the three detection methods (a–c). (d) $M \pm SD$ Δppm as determined by CRDS (•), IRMS-WT (■), and IRMS-PLA (▽). *Significantly different ($p < .05$) from 0 min. +Significantly different ($p < .05$) from 45 min, $n = 8$. D_2O = deuterium oxide; CRDS = cavity ring-down spectroscopy; IRMS = isotope-ratio mass spectrometry; WT = equilibrated water; PLA = plasma.

methods. There were also no model ($F = 0.001$, $df = 1$, $p = .975$, $\eta^2 = .00$) differences (TRI: 11.73 ± 4.7 min; SIG: 11.76 ± 3.3) or Model × Method interaction effects ($F = 1.88$, $df = 2$, $p = .19$, $\eta^2 = .24$). One outlier with extremely slow linear D_2O (solid flat line at bottom, Figure 2a–2c) was not included since $t_{1/2max}$ was unable to be fit by SIG.

Two distinct D_2O appearance profiles (slow and fast individuals) and the corresponding derivation of $t_{1/2max}$ are plotted for SIG and TRI. For SIG, a "slow" rate individual exhibiting a roughly sigmoid time course (Figure 3a) is compared with an individual with "fast" rapid rise followed by a decline (Figure 3c), respectively. The vertical lines indicate $t_{1/2max}$ derived from Equation (2). Dashed lines in Figure 3c indicate the sigmoid fit and $t_{1/2max}$ for the entire time course including the declining phase, while solid lines indicate the fit restricted to the rising phase and $t_{1/2max}$ derived from this fit. In Figure 3b and 3d, $t_{1/2max}$ by TRI for "slow" versus "fast" individuals, respectively, are plotted against time along with derivatives T_1 and T_2 (Figure 3b and 3d, insets). Appearance rates for both "slow" and "fast" profiles achieve identifiable peaks (T_1). T_2 is the time at which the rate most closely approaches zero. The effect of using T_2 to calculate $t_{1/2max}$ (for TRI) when the appearance rate does not reach zero is unclear (Figure 3b, inset) since the plateau might have mathematically occurred either between 30 and 45 or after 45 min. These individual data suggested SIG and TRI may yield different $t_{1/2max}$ values.

D_2O appearance with CRDS ($n = 14$) was used to compare $t_{1/2max}$ models. SIG and TRI were highly correlated (Figure 4a) within acceptable (1.96 *SD*) limits of agreement (Figure 4b). Mean bias was 0.95 min, but the slope of the regression line (Figure 4b) differed from zero ($p = .029$), indicating a greater discrepancy between models as the relative hydration speed became slower (higher $t_{1/2max}$) across individuals. Two subgroups were stratified based on fluid delivery speed above and below the $t_{1/2max}$ grand mean (i.e., 12 min). Effects for $t_{1/2max}$ model ($F = 9.29$, $df = 1$, $p = .01$, $\eta^2 = .44$) and Model × Speed Group interaction ($F = 12.1$, $df = 1$, $p = .005$, $\eta^2 = .50$) were significant (Figure 5).

The $t_{1/2max}$ differential (5.8 min) between "fast" ($n = 8$) and "slow" ($n = 6$) subgroups was significant ($F = 33.0$, $df = 1$, $p < .001$, $\eta^2 = .73$). The $t_{1/2max}$ model used for "slow" (TRI: 16.1 ± 1.7 min; 95% confidence interval [14.5, 17.7]) versus SIG (13.7 ± 2.1 min; 95% confidence interval [11.8, 15.5]) yielded a speed difference ($p = .001$), unlike the "fast" subgroup (TRI: 9.0 ± 1.9 min vs. SIG: 9.2 ± 2.1 min, $p = .75$). Thus, the model selected to calculate $t_{1/2max}$ made no difference in "fast" individuals, but TRI yielded a significantly "slower" $t_{1/2max}$ than SIG in "slow" individuals.

Discussion

The appearance of D_2O in blood has long been used to compare the effects of beverage composition and conditions (rest vs. exercise) on relative fluid delivery (Davis et al., 1987; Jeukendrup et al., 2009). Initial studies (Davis et al., 1990; Gisolfi et al., 1990; Lambert et al., 1999) utilized infrared spectroscopy, which required

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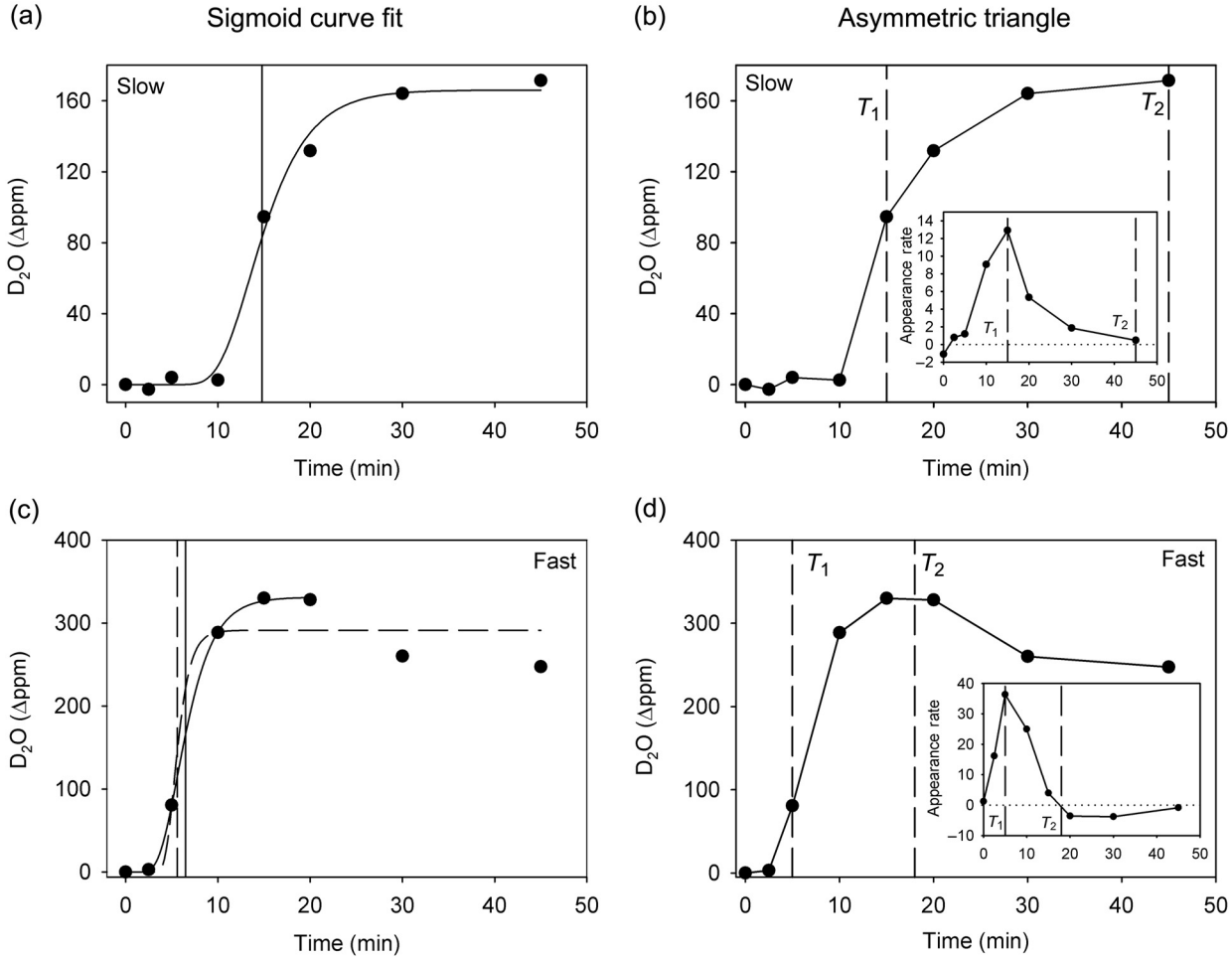

Figure 3 — Determination of relative fluid delivery speed ($t_{1/2max}$) based on two models: sigmoid curve fitting (a and c) and asymmetric triangle (b and d) when individual profiles are "slow" and "fast." (a, Slow): A sigmoid curve fits the plasma D_2O profile, $r^2 = .993$. The solid vertical line is the halftime, $t_{1/2max} = 14.75$ min. (c, Fast): A sigmoid curve yielded a poorer fit, $R^2 = .958$, to the profile (dashed curve). The dashed vertical line is the $t_{1/2max} = 5.59$ min. Fitting only the rising phase of the D_2O profile improved the fit, $r^2 > .999$ (solid curve) but increased the $t_{1/2max} = 6.49$ min (solid vertical line). Time course of plasma D_2O appearance for the same slow (b) and fast (d) individual shown with derivations from asymmetric triangle. Insets show rate of the plasma D_2O appearance. Vertical lines in the both main and inset panels indicate the parameters T_1 and T_2. Slow $t_{1/2max} = 15.00$ min and fast $t_{1/2max} = 6.71$ min. D_2O = deuterium oxide; $t_{1/2max}$ = time to half-maximal.

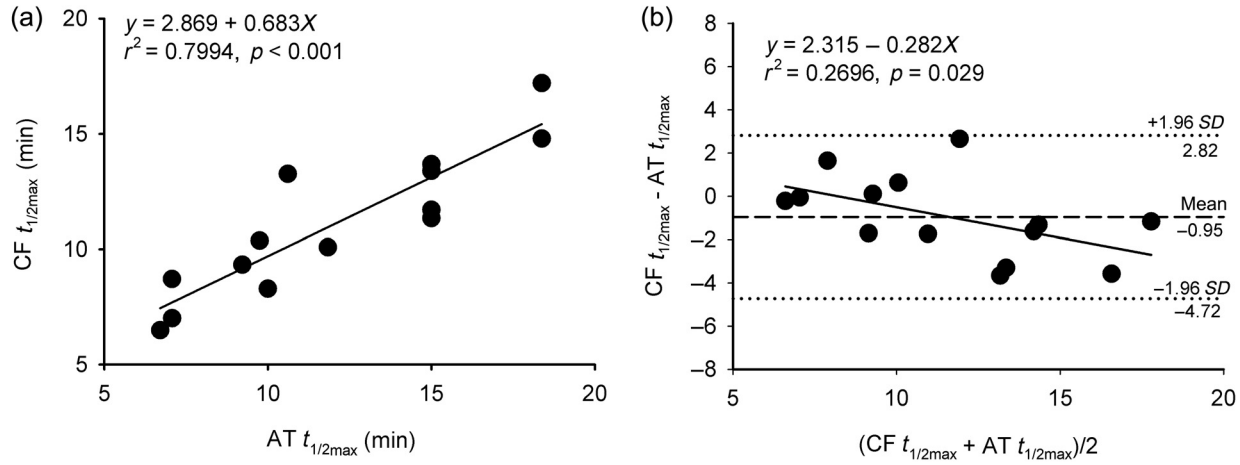

Figure 4 — Linear regression (a) and Bland–Atman plot (b) of D_2O appearance halftime maximum derived from AT model or sigmoid CF ($n = 14$ participants). Solid line in (a) represents linear regression line of best fit, $R^2 = .799$. (b) Dashed line is the AT bias, 0.95 min; the dotted lines represent the 1.96 *SD* limit of agreement (2.82; –4.72 min). Solid line is the linear regression line of best fit, $R^2 = .270$, and slope is significantly different from zero, $p = .029$. AT = asymmetric triangle; CF = curve fitting; D_2O = deuterium oxide; $t_{1/2max}$ = time to half-maximal.

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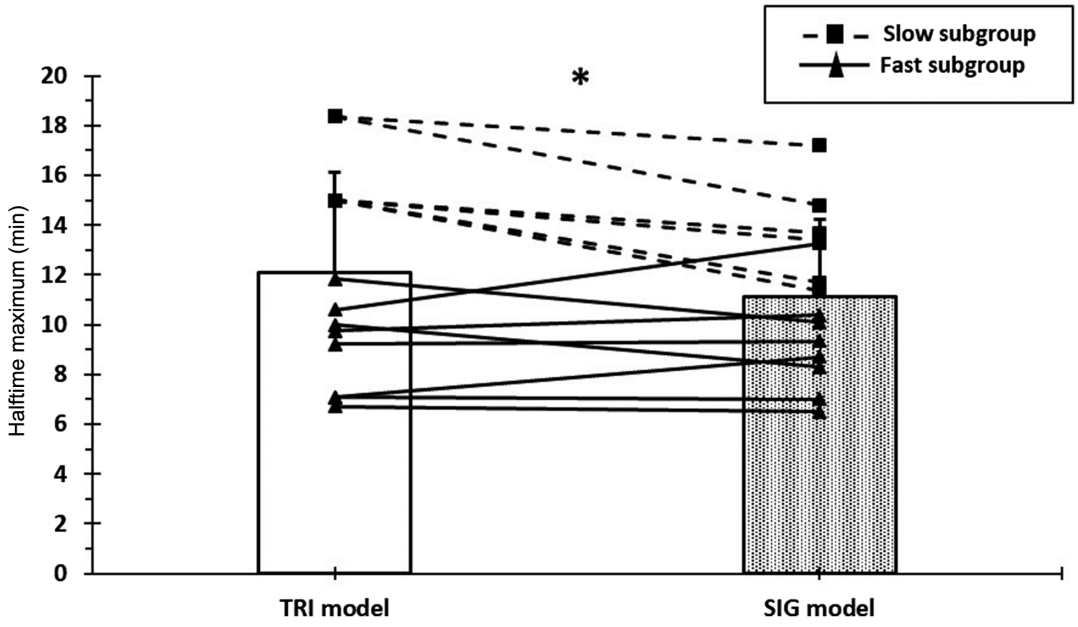

Figure 5 — Mean ± *SD* D_2O halftime maximum derived by TRI and SIG models according to subgroups of individuals exhibiting "fast" (solid lines with ▲) and "slow" (dashed line with ■) speed profiles (categorized by below and above 12 min). *Significant model effect, Model × Speed Group interaction effect, and subgroup differences ($p < .05$). D_2O = deuterium oxide; TRI = asymmetric triangle; SIG = sigmoid curve fit.

distillation of water from plasma. This method gave way to IRMS (Jeukendrup et al., 2009; Koulmann et al., 1997; Maughan et al., 1990; Millard-Stafford et al., 2007; Péronnet et al., 2012; Rehrer et al., 1992; Rowlands et al., 2011), which required vapor equilibration or distillation to fractionate D_2O from plasma. Although CRDS has been used to analyze urine for doubly labeled water experiments (Berman et al., 2015; Sagayama et al., 2021), this is the first study to verify CRDS for plasma-based oral solution bioavailability. CRDS produced values not statistically different compared with IRMS (with two distinct plasma equilibration and fractionation methods) across the array of beverage bioavailability parameters (area under the curve, time to peak, $t_{1/2max}$, and accumulation rates), although up to 20% of the variability was attributed to the laboratory method used. Comparisons fell within acceptable levels of agreement (93%–95%) with a mean bias within −12 ppm for CRDS compared to IRMS. However, the two models used to calculate relative speed of fluid delivery were affected twofold more by the influence of interindividual variability (≥40%–50% of variance) in the rate of appearance in individual D_2O profiles.

Despite measured background [D_2O] differences (and significant underreporting for CRDS), there was no significant impact on D_2O appearance curves based on method. This is due to the essential step of utilizing net Δppm (i.e., absolute [D_2O] baseline values are subtracted), since prior D_2O administration (half-life ~10 days) increases background when participants undergo successive experimental trials. Distilled water has a natural D_2O background level of 140 ppm (Davis et al., 1987). However, population variations in natural background D_2O are well documented based on geographic location, water supply (Horvitz & Schoeller, 2001), and seasonal changes (Harbison et al., 2015; Nordøy et al., 2017), in addition to individual differences over time (Berman et al., 2015). While variability in background [D_2O] introduces error into doubly labeled water measures calculating total energy expenditure, background [D_2O] variability has a minor impact on fluid delivery measures. Therefore, biomarkers of beverage bioavailability are robust and less influenced by laboratory methods differing by sample preparation or technology, indicating CRDS is a suitable method for plasma-based oral hydration studies.

Another notable finding was that two different models for calculating speed of fluid delivery were not interchangeable. Therefore, individual variability should dictate which model to use for derivation of $t_{1/2max}$. Given the agreement among methods (IRMS and CRDS; Figure 2d), the majority of this variability must be attributed to individual physiological differences in rates of gastric emptying (Maughan & Leiper, 1996), intestinal absorption, and/or speed of absorbed water entering body pools (Leiper, 2015). For example, considerable interindividual variability for intestinal absorption as high as 22% was documented (Leiper et al., 1992) but within-individual variability relatively stable (CV between 3% and 12%). In the present study, the appearance phase was complete in one participant by 20 min (D_2O plateau followed by a decline), while, in others, there was no apparent plateau. Furthermore, there may be "outliers" (bottom trace, Figure 2a–2c) whose appearance rate is linear with a gradual slope. Since this individual had similar profiles across all three independent laboratory methods, the slow time course cannot be attributed to measurement artifact. Here, additional considerations are suggested for adopting the optimal analytical model to calculate relative speed of fluid delivery.

Two different pharmacokinetic D_2O profiles were observed, similar to a previous study (Péronnet et al., 2012). Few other studies display the individual variability inherent in D_2O appearance rates. In the present study, one subgroup exhibited relatively "slow" fluid delivery characterized only by a rising phase of D_2O. Data from "slow" individuals were well fit (average $r^2 = .989$) by SIG. Since the rate of D_2O appearance never reached zero, the T_2

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parameter required for TRI derivation was influenced by an indeterminate amount. In these cases, the nonlinear curve-fitting approach (e.g., SIG) appears preferable despite appearing to result in faster $t_{1/2max}$ versus TRI. In contrast, TRI may be more appropriate for "fast absorbers" when plasma [D_2O] either declines after reaching a peak or achieves a mathematical plateau, although no meaningful group difference was observed for "fast" individuals for TRI compared with SIG. In one case, SIG did not satisfactorily fit a "fast" initial rapid increase in D_2O followed by decline after plateau. This physiological profile was not adequately described by a simple sigmoid curve, and thus, TRI appears to be the preferred approach (Bluck et al., 2002; Hill et al., 2004) to calculate $t_{1/2max}$. Although TRI was validated for oral hydration studies (Hill et al., 2008), the conditions establishing its validity were not clearly defined. It is also possible that there are exceptions to these recommendations since the exceptionally slow linear response in one outlier was not adequately fit by either model, thus limiting the ability to calculate $t_{1/2max}$ with confidence in all individuals.

In the present study, the 6 min time differential between "slow versus fast" individuals had practical significance, particularly in athletic settings when relatively short intermission periods (e.g., 15 min in FIFA football) represent the main opportunity for fluid replacement. In terms of personalized nutrition, "slow"-absorbing athletes competing in thermally challenging endurance events might be better served with lower carbohydrate beverages as evidenced by the 6 min interbeverage difference in $t_{1/2max}$ previously reported between 3% and 6% carbohydrate solutions (Jeukendrup et al., 2009). Identification of such individuals could be beneficial in competitive sports where fluid delivery speed has greater importance, although the cost benefit of such analysis is likely unrealistic.

This study is not without limitations. Our study appears underpowered to differentiate among laboratory methods given complete data were only available for eight participants. Additional testing was limited by the financial burden coupled with retirement of the IRMS (and technician) during the COVID-19 pandemic. The research design may have influenced $t_{1/2max}$ results. A longer blood sampling time or more frequent samples between 30 and 45 might have provided definitive plateaus (yielding the required derivative of 0 for T_2 when using TRI) for "slow" absorbers. However, previous comparable studies reported mean plasma D_2O peaking before 45 min. Hypotonic beverages peaked rapidly and were already lower by 50 min (Currell et al., 2008). During rest, D_2O peaked between 10 and 20 min and fell by 25 min (while exercise delayed the plateau; Maughan et al., 1990). D_2O appearance plateaued at 30 and was lower by 45 min in 0% glucose solutions (while adding carbohydrate delayed the peak at rest; Jeukendrup et al., 2009). Thus, for studies examining different beverage formulations and/or under exercise conditions, additional sampling points beyond that used in the present study are indicated.

In summary, CRDS and IRMS produce similar conclusions for plasma D_2O detection when assessing fluid bioavailability biomarkers. Despite the additional plasma preparation steps for IRMS and measured differences in background [D_2O] in CRDS, beverage bioavailability biomarkers appear remarkably robust across laboratory techniques. Therefore, the choice of spectroscopic method may be driven by cost and availability. However, the large interindividual biological variation in D_2O appearance curves should factor into the choice of the most appropriate model for calculating speed of fluid delivery. In particular, the TRI model may not be appropriate for individuals with slow fluid delivery speeds.

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References

Armstrong, L.E., Klau, J.F., Ganio, M.S., McDermott, B.P., Yeargin, S.W., Lee, E.C., & Maresh, C.M. (2010). Accumulation of ^2H_2O in plasma and eccrine sweat during exercise-heat stress. *European Journal of Applied Physiology, 108*(3), 477–482. https://doi.org/10.1007/s00421-009-1223-2

Armstrong, L.E., Pumerantz, A.C., Fiala, K.A., Roti, M.W., Kavouras, S.A., Casa, D.J., & Maresh, C.M. (2010). Human hydration indices: Acute and longitudinal reference values. *International Journal of Sport Nutrition and Exercise Metabolism, 20*(2), 145–153. https://doi.org/10.1123/ijsnem.20.2.145

Berman, E.S., Fortson, S.L., Snaith, S.P., Gupta, M., Baer, D.S., Chery, I., Blanc, S., Melanson, E.L., Thomson, P.J., & Speakman, J.R. (2012). Direct analysis of δ^2H and $\delta^{18}O$ in natural and enriched human urine using laser-based, off-axis integrated cavity output spectroscopy. *Analytical Chemistry, 84*(22), 9768–9773. https://doi.org/10.1021/ac3016642

Berman, E.S., Melanson, E.L., Swibas, T., Snaith, S.P., & Speakman, J.R. (2015). Inter- and intraindividual correlations of background abundances of ^2H, ^{18}O and ^{17}O in human urine and implications for DLW measurements. *European Journal of Clinical Nutrition, 69*(10), 1091–1098. https://doi.org/10.1038/ejcn.2015.10

Bland, J.M., & Altman, D.G. (1986). Statistical methods for assessing agreement between two methods of clinical measurement. *Lancet, 1*(8476), 307–310.

Bluck, L.J., Harding, M., French, S., Wright, A., Halliday, D., & Coward, W.A. (2002). Measurement of gastric emptying in man using deuterated octanoic acid. *Rapid Communications in Mass Spectrometry, 16*(2), 127–133. https://doi.org/10.1002/rcm.541

Church, A., Lee, F., & Buono, M.J. (2017). Transition duration of ingested deuterium oxide to eccrine sweat during exercise in the heat. *Journal of Thermal Biology, 63,* 88–91. https://doi.org/10.1016/j.jtherbio.2016.11.018

Currell, K., Urch, J., Cerri, E., Jentjens, R.L., Blannin, A.K., & Jeukendrup, A.E. (2008). Plasma deuterium oxide accumulation following ingestion of different carbohydrate beverages. *Applied Physiology, Nutrition, and Metabolism, 33*(6), 1067–1072. https://doi.org/10.1139/H08-084

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Damas, F., Phillips, S.M., Libardi, C.A., Vechin, F.C., Lixandrão, M.E., Jannig, P.R., Costa, L.A., Bacurau, A.V., Snijders, T., Parise, G., Tricoli, V., Roschel, H., & Ugrinowitsch, C. (2016). Resistance training-induced changes in integrated myofibrillar protein synthesis are related to hypertrophy only after attenuation of muscle damage. *Journal of Physiology, 594*(18), 5209–5222. https://doi.org/10.1113/jp272472

Davis, J.M., Burgess, W.A., Slentz, C.A., & Bartoli, W.P. (1990). Fluid availability of sports drinks differing in carbohydrate type and concentration. *American Journal of Clinical Nutrition, 51*(6), 1054–1057. https://doi.org/10.1093/ajcn/51.6.1054

Davis, J.M., Lamb, D.R., Burgess, W.A., & Bartoli, W.P. (1987). Accumulation of deuterium oxide in body fluids after ingestion of D_2O-labeled beverages. *Journal of Applied Physiology, 63*(5), 2060–2066. https://doi.org/10.1152/jappl.1987.63.5.2060

Davis, J.M., Lamb, D.R., Pate, R.R., Slentz, C.A., Burgess, W.A., & Bartoli, W.P. (1988). Carbohydrate-electrolyte drinks: Effects on endurance cycling in the heat. *American Journal of Clinical Nutrition, 48*(4), 1023–1030. https://doi.org/10.1093/ajcn/48.4.1023

dos Santos, T.H.R., do Rosário Zucchi, M., Lemaire, T.J., de Azevedo, A.E.G., & Viola, D.N. (2019). A statistical analysis of IRMS and CRDS methods in isotopic ratios of $^2H/^1H$ and $^{18}O/^{16}O$ in water. *SN Applied Sciences, 1*(7), Article 664. https://doi.org/10.1007/s42452-019-0635-9

Giavarina, D. (2015). Understanding Bland Altman analysis. *Biochemia Medica, 25*(2), 141–151. https://doi.org/10.11613/bm.2015.015

Gisolfi, C.V., Summers, R.W., Schedl, H.P., Bleiler, T.L., & Oppliger, R.A. (1990). Human intestinal water absorption: Direct vs. indirect measurements. *American Journal of Physiology, 258*(2, Pt. 1), G216–G222. https://doi.org/10.1152/ajpgi.1990.258.2.G216

Harbison, J.E., Dugas, L.R., Brieger, W., Tayo, B.O., Alabi, T., Schoeller, D.A., & Luke, A. (2015). Seasonal variation in natural abundance of ^2H and ^{18}O in urine samples from rural Nigeria. *Journal of Applied Physiology, 119*(1), 55–60. https://doi.org/10.1152/japplphysiol.00894.2014

Hill, R.J., Bluck, L.J., & Davies, P.S. (2004). Using a non-invasive stable isotope tracer to measure the absorption of water in humans. *Rapid Communications in Mass Spectrometry, 18*(6), 701–706. https://doi.org/10.1002/rcm.1391

Hill, R.J., Bluck, L.J., & Davies, P.S. (2008). The hydration ability of three commercially available sports drinks and water. *Journal of Science and Medicine in Sport, 11*(2), 116–123. https://doi.org/10.1016/j.jsams.2006.12.117

Horvitz, M.A., & Schoeller, D.A. (2001). Natural abundance deuterium and 18-oxygen effects on the precision of the doubly labeled water method. *American Journal of Physiology: Endocrinology and Metabolism, 280*(6), E965–E972. https://doi.org/10.1152/ajpendo.2001.280.6.E965

Jackson, A.S., & Pollock, M.L. (1978). Generalized equations for predicting body density of men. *British Journal of Nutrition, 40*(3), 497–504. https://doi.org/10.1079/bjn19780152

Jeukendrup, A.E., Currell, K., Clarke, J., Cole, J., & Blannin, A.K. (2009). Effect of beverage glucose and sodium content on fluid delivery. *Nutrition and Metabolism, 6,* Article 9. https://doi.org/10.1186/1743-7075-6-9

Jeukendrup, A.E., & Moseley, L. (2010). Multiple transportable carbohydrates enhance gastric emptying and fluid delivery. *Scandanavian Journal of Medicine and Science in Sports, 20*(1), 112–121. https://doi.org/10.1111/j.1600-0838.2008.00862.x

Johannsen, D.L., Knuth, N.D., Huizenga, R., Rood, J.C., Ravussin, E., & Hall, K.D. (2012). Metabolic slowing with massive weight loss despite preservation of fat-free mass. *Journal of Clinical Endocrinology and Metabolism, 97*(7), 2489–2496. https://doi.org/10.1210/jc.2012-1444

Koulmann, N., Melin, B., Jimenez, C., Charpenet, A., Savourey, G., & Bittel, J. (1997). Effects of different carbohydrate-electrolyte beverages on the appearance of ingested deuterium in body fluids during moderate exercise by humans in the heat. *European Journal of Applied Physiology and Occupation Physiology, 75*(6), 525–531. https://doi.org/10.1007/s004210050199

Lambert, C.P., Ball, D., Leiper, J.B., & Maughan, R.J. (1999). The use of a deuterium tracer technique to follow the fate of fluids ingested by human subjects: Effects of drink volume and tracer concentration and content. *Experimental Physiology, 84*(2), 391–399.

Leiper, J.B. (2015). Fate of ingested fluids: Factors affecting gastric emptying and intestinal absorption of beverages in humans. *Nutrition Reviews, 73,* 57–72. https://doi.org/10.1093/nutrit/nuv032

Leiper, J.B., Brouns, F., Maughan, R.J. (1992). Absorption of water and solute from sucrose electrolyte solutions containing calcium and magnesium. *Proceedings of the Nutrition Society, 51,* Article 142.

Lenhard, W., & Lenhard, A. (2016). Computation of effect sizes. *Psychometrica*. https://www.psychometrica.de/effect_size.html

Lukaski, H.C., & Johnson, P.E. (1985). A simple, inexpensive method of determining total body water using a tracer dose of D_2O and infrared absorption of biological fluids. *American Journal of Clinical Nutrition, 41*(2), 363–370. https://doi.org/10.1093/ajcn/41.2.363

Maughan, R.J, & Leiper, J.B. (1996). Methods for the assessment of gastric emptying in humans: An overview. *Diabetic Medicine, 13*(Suppl. 9), S6–S10.

Maughan, R.J., Leiper, J.B., & McGaw, B.A. (1990). Effects of exercise intensity on absorption of ingested fluids in man. *Experimental Physiology, 75*(3), 419–421. https://doi.org/10.1113/expphysiol.1990.sp003419

Maughan, R.J., Leiper, J.B., & Vist, G.E. (2004). Gastric emptying and fluid availability after ingestion of glucose and soy protein hydrolysate solutions in man. *Experimental Physiology, 89*(1), 101–108. https://doi.org/10.1113/expphysiol.2003.002655

McKinney C., McCrea J., Epstein S., Allen, H.A., & Urey, H.C. (1950). Improvements in mass spectrometers for the measurement of small differences in isotope abundance ratios. *The Review of Scientific Instruments, 21*(8), 724–730. https://doi.org/10.1063/1.1745698

Millard-Stafford, M.L., Cureton, K.J., Wingo, J.E., Trilk, J., Warren, G.L., & Buyckx, M. (2007). Hydration during exercise in warm, humid conditions: Effect of a caffeinated sports drink. *International Journal of Sport Nutrition and Exercise Metabolism, 17*(2), 163–177. https://doi.org/10.1123/ijsnem.17.2.163

Nordøy, E.S., Lager, A.R., & Schots, P.C. (2017). Seasonal changes in background levels of deuterium and oxygen-18 prove water drinking by harp seals, which affects the use of the doubly labelled water method. *The Journal of Experimental Biology, 220*(Pt. 23), 4450–4455. https://doi.org/10.1242/jeb.161943

Penna, D., Stenni, B., Šanda, M., Wrede, S., Bogaard, T.A., Michelini, M., . . . Wassenaar, L.I. (2012). Technical note: Evaluation of between-sample memory effects in the analysis of δ^2H and $\delta^{18}O$ of water samples measured by laser spectroscopes. *Hydrology Earth Systems Science, 16*(10), 3925–3933. https://doi.org/10.5194/hess-16-3925-2012

Péronnet, F., Mignault, D., du Souich, P., Vergne, S., Le Bellego, L., Jimenez, L., & Rabasa-Lhoret, R. (2012). Pharmacokinetic analysis of absorption, distribution and disappearance of ingested water labeled with D_2O in humans. *European Journal of Applied Physiology, 112*(6), 2213–2222. https://doi.org/10.1007/s00421-011-2194-7

Rehrer, N.J., Wagenmakers, A.J., Beckers, E.J., Halliday, D., Leiper, J.B., Brouns, F., Maughan, R.J., Westerterp, K., & Saris, W.H. (1992). Gastric emptying, absorption, and carbohydrate oxidation during prolonged exercise. *Journal of Applied Physiology, 72*(2), 468–475. https://doi.org/10.1152/jappl.1992.72.2.468

Rowlands, D.S., Bonetti, D.L., & Hopkins, W.G. (2011). Unilateral fluid absorption and effects on peak power after ingestion of commercially

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available hypotonic, isotonic, and hypertonic sports drinks. *International Journal of Sport Nutrition and Exercise Metabolism, 21*(6), 480–491. https://doi.org/10.1123/ijsnem.21.6.480

Sagayama, H., Racine, N.M., Shriver, T.C., & Schoeller, D.A. (2021). Comparison of isotope ratio mass spectrometry and cavity ring-down spectroscopy procedures and precision of the doubly labeled water method in different physiological specimens. *Rapid Communications in Mass Spectrometry, 35*(21), Article 9188. https://doi.org/10.1002/rcm.9188

Thorsen, T., Shriver, T., Racine, N., Richman, B.A., & Schoeller, D.A. (2011). Doubly labeled water analysis using cavity ring-down spectroscopy. *Rapid Communications in Mass Spectrometry, 25*(1), 3–8. https://doi.org/10.1002/rcm.4795

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